Google Cloud

Introduction to AI and Machine Learning on Google Cloud



Module 4

AI development workflow

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01 AI foundations ☑

02 Generative AI ☑

03 AI development options ☑

04 AI development workflow ☐

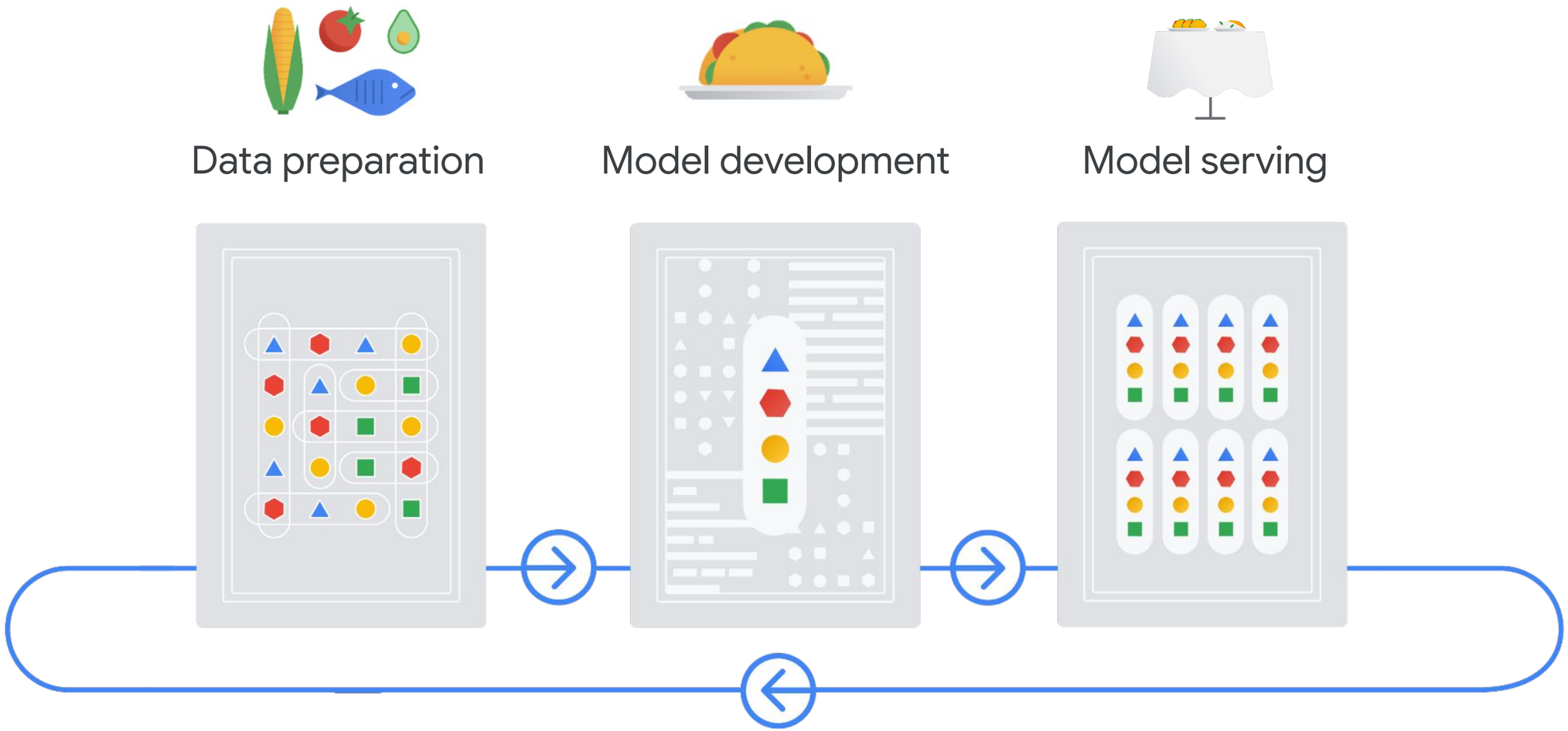
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ML workflow

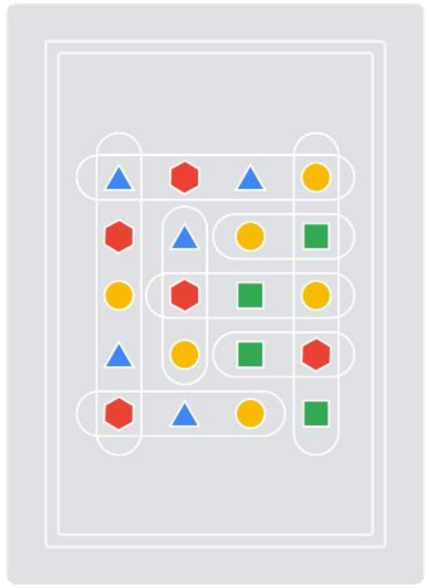
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Machine learning workflow



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Stage 1: Data preparation



Two steps

- Upload data
- Engineer features

Data types

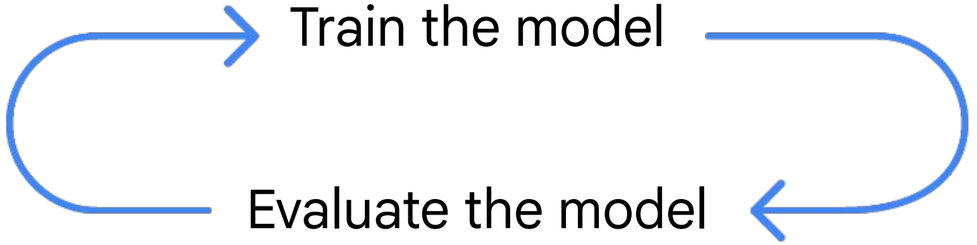
- Streaming versus batch data
- Structured versus unstructured data

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Stage 2: Model development

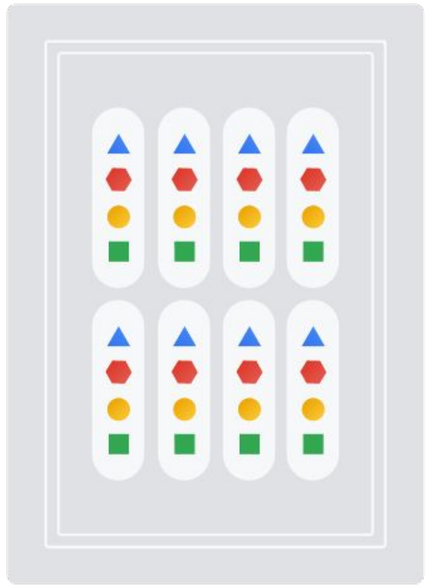


A model needs a tremendous amount of iterative training.



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Stage 3: Model serving



A model needs to actually be used in order to predict results.

- Deploy the model.
- Monitor the model.

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ML workflow is iterative

Prepare raw ingredients.

Data preparation

Experiment with recipes.

Model development

Serve the meal.

Model serving



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No-code versus code-based option



Agent Platform AutoML

- No-code approach
- Configuration through UI
- User friendly



Agent Platform Workbench or Agent Platform Colab Enterprise

- Code-based approach
- SDKs included
- Experienced users

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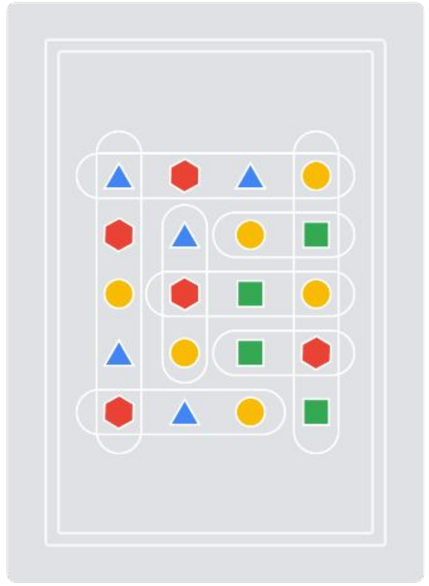


Data preparation

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Data preparation

Data preparation



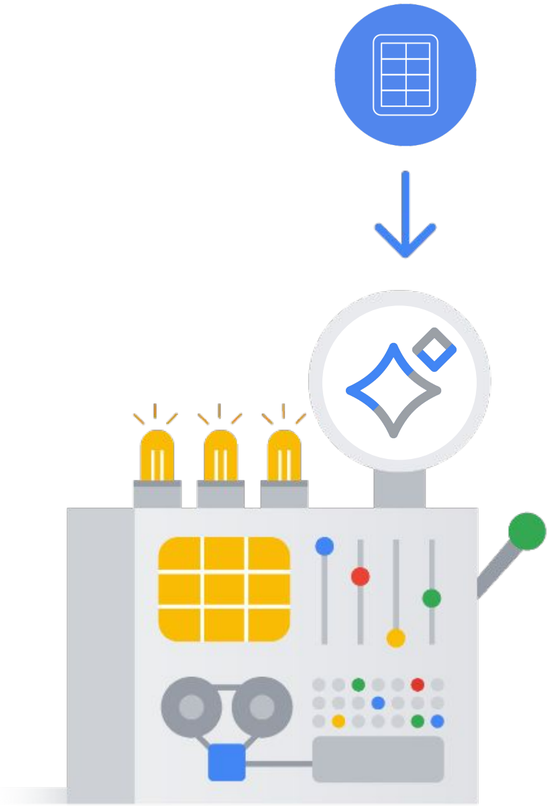
Prepare raw ingredients.

01 Upload data.

02 Engineer features.

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Upload data: Select data sources



Cloud Storage

BigQuery

Local machine

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Upload data: Decide objectives



Tabular data



AutoML solves different types of problems, called **objectives**.

Regression or classification

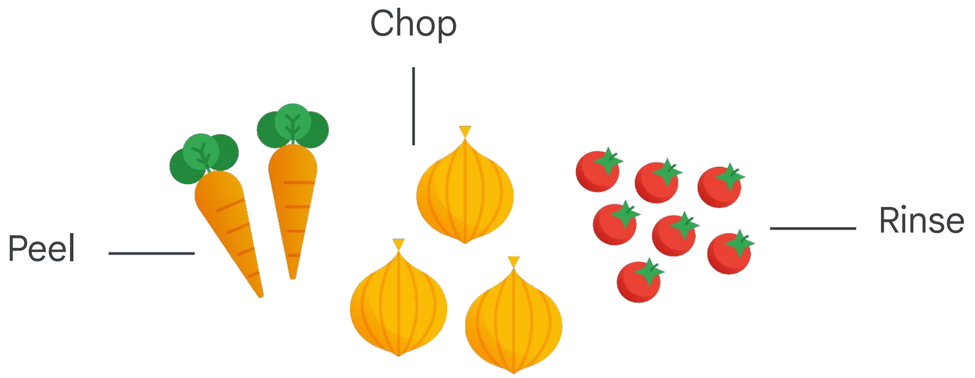
Forecasting

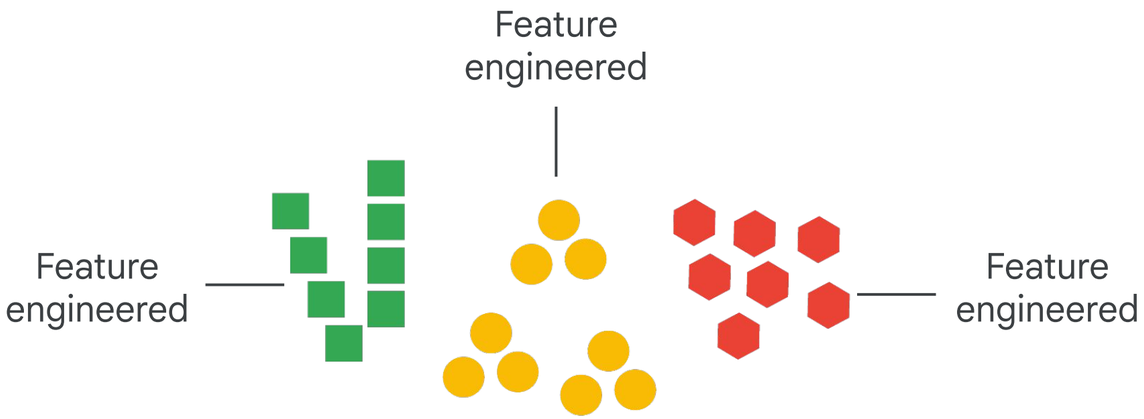
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Feature engineering

A feature refers to a factor that contributes to the prediction.

- An independent variable
- A column in a table





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Agent Platform Feature Store



Agent Platform Feature Store

A centralized repository to manage, serve, and share features

- Real-time (online) serving.
- Batch (offline) serving.

At scale, low latency

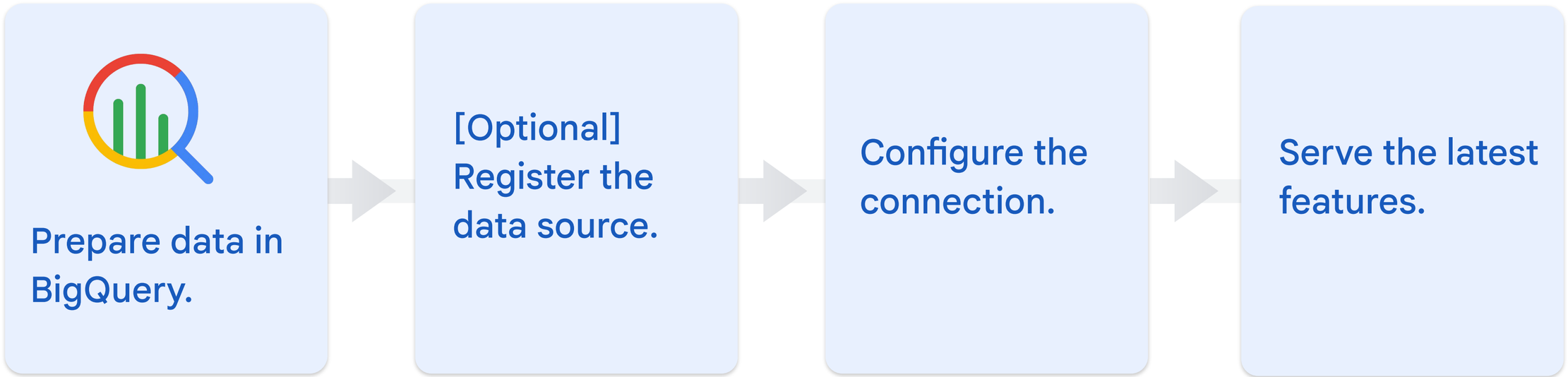
Ready for generative AI

- Manage and serve embeddings.
- Retrieve similar items.

In real time

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Agent Platform Feature Store workflow



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Benefits of Agent Platform Feature Store

- Features are shareable for training and serving.
- Features are reusable.
- Features are scalable.
- Features are easy to use.

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Model development

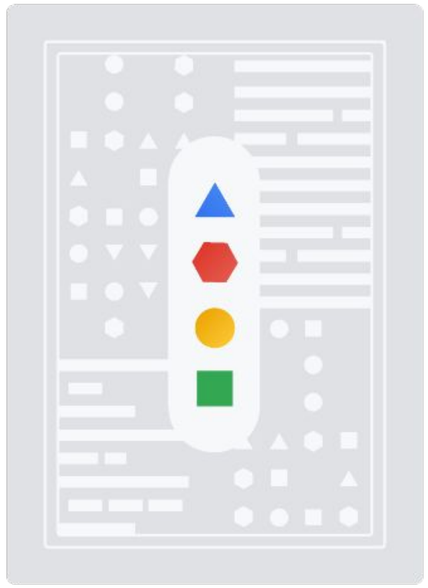
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Model development

Data preparation

Prepare raw ingredients.

Model development



Experiment with recipes.

01 Train model.

02 Evaluate model.

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Train new model

1. Training method
2. Model details
3. Training options
4. Compute and pricing

START TRAINING CANCEL

Dataset *

retailSales

Objective *

Classification

Model training method

- (•) Agent Platform AutoML
- ( ) Custom training (advanced)

CONTINUE

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Train new model

1. Training method
2. Model details
3. Training options
4. Compute and pricing

START TRAINING CANCEL

Train new model

Train new version

Name *

retailSales

Description

Target column *

ADVANCED OPTIONS

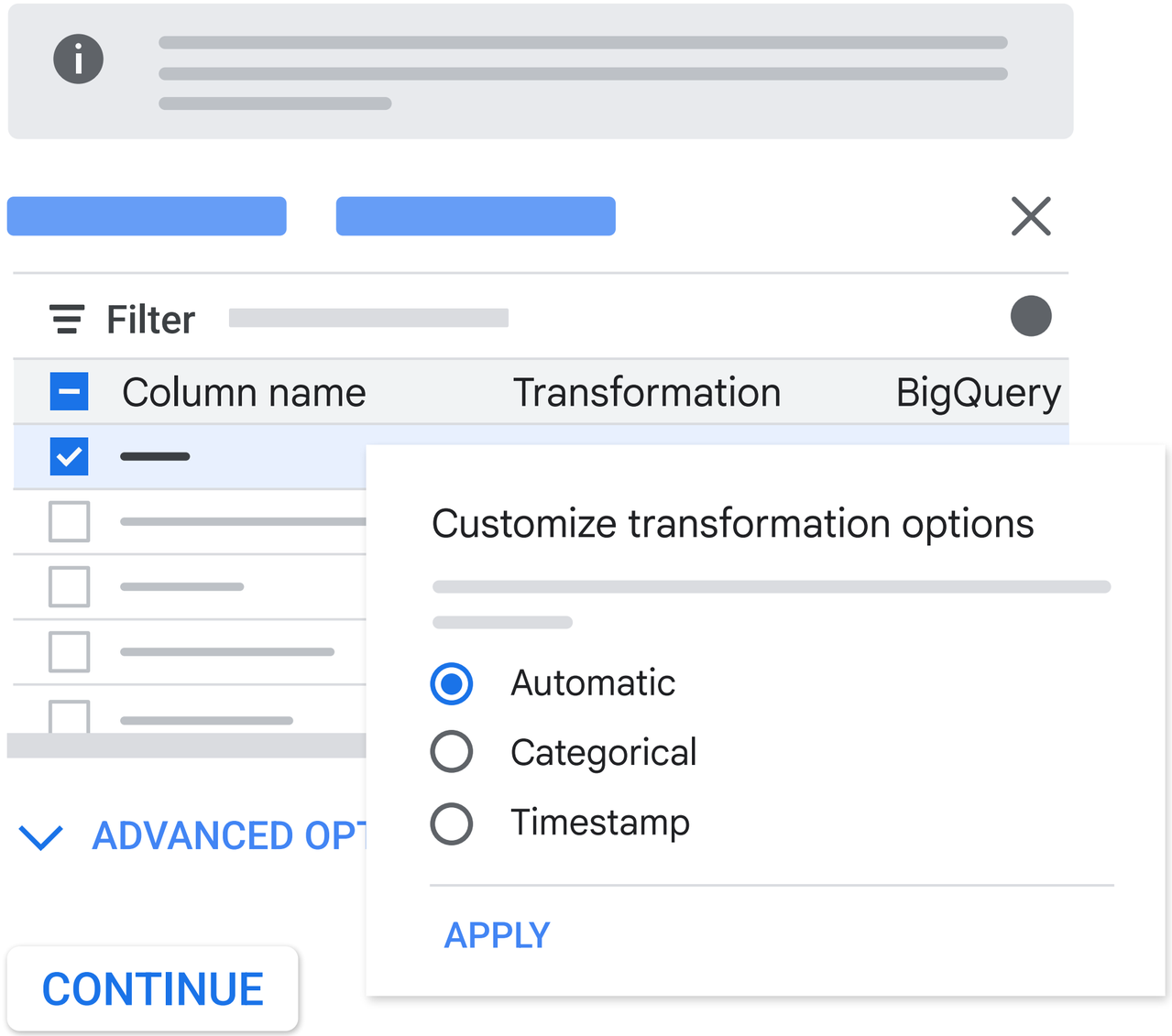
CONTINUE

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Train new model

- ✓ Training method
- ✓ Model details
- 3 Training options
- 4 Compute and pricing

START TRAINING CANCEL



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Train new model

- ✓ Training method
- ✓ Model details
- ✓ Training options
- 4 Compute and pricing

START TRAINING CANCEL



Budget *

1 Maximum node hours

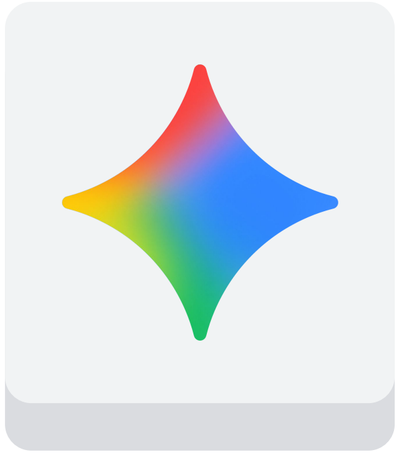
Estimated completion: May 4, 2023 1 PM GMT+2

✓ Enable early stopping

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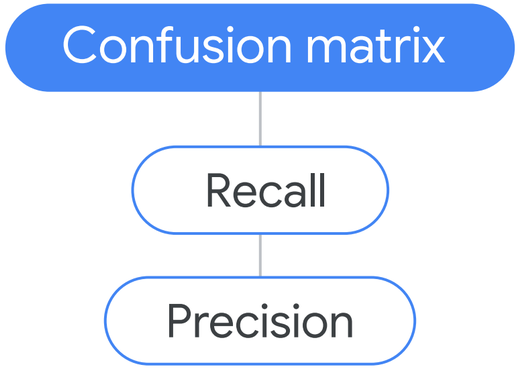
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Evaluation model: Metrics



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Evaluation metrics



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Confusion matrix

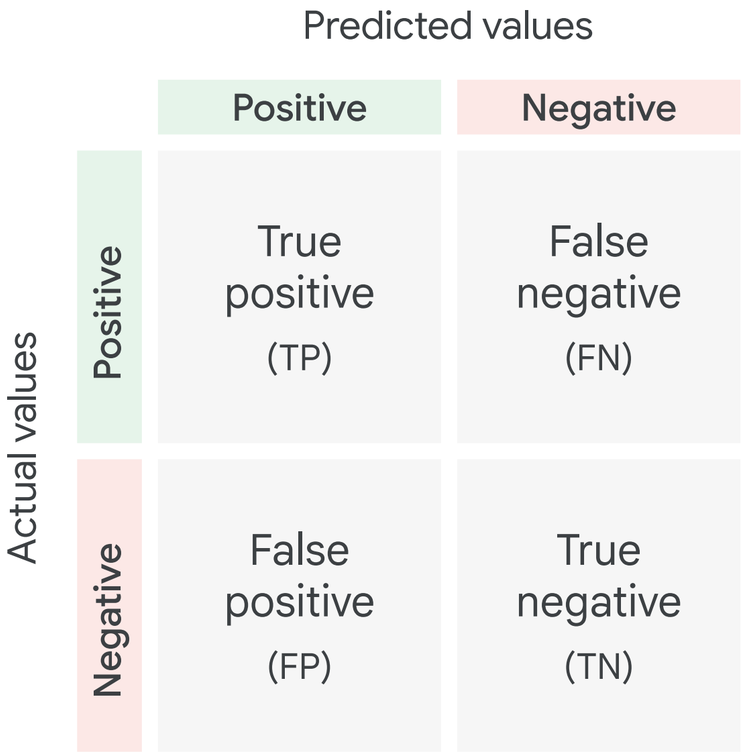
A performance measurement for machine learning classification problems

Actual values \ Predicted values	Positive (cat)	Negative (not cat)
Positive (cat)	True positive  This is a cat.	False negative  This is not a cat. Type 2 error
Negative (not cat)	False positive  This is a cat. Type 1 error	True negative  This is not a cat.

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Recall and precision



$$\text{Recall} = \frac{TP}{TP+FN}$$

Recall

Refers to all the positive cases, and looks at how many were predicted correctly.

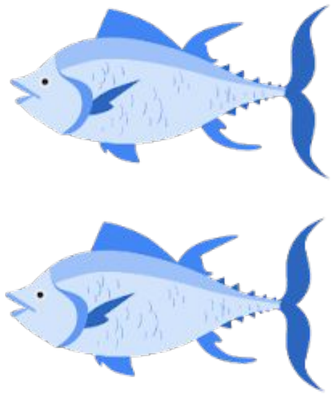
Precision

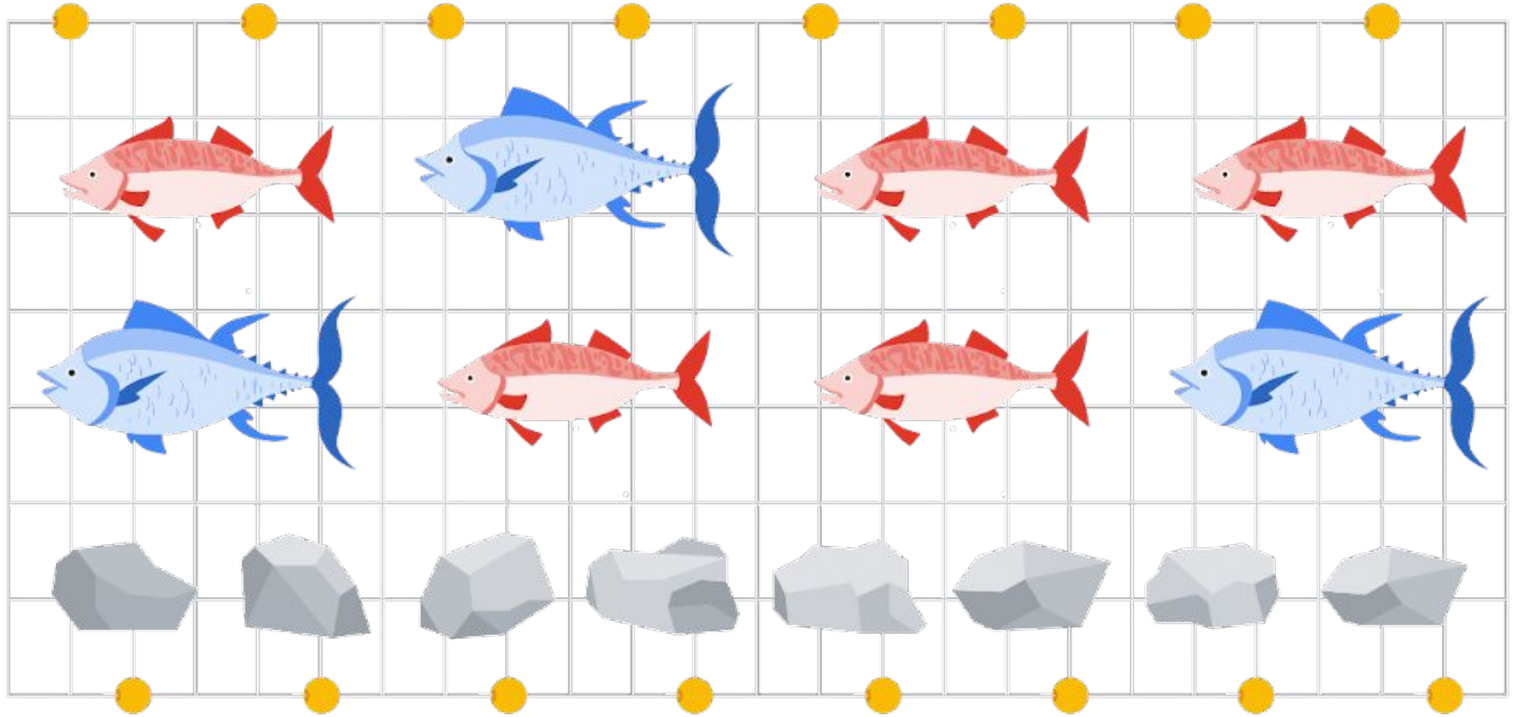
Refers to all the cases predicted as positive, and how many are actually positive.

$$\text{Precision} = \frac{TP}{TP+FP}$$

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Example: Fishing with a wide net





Wide net

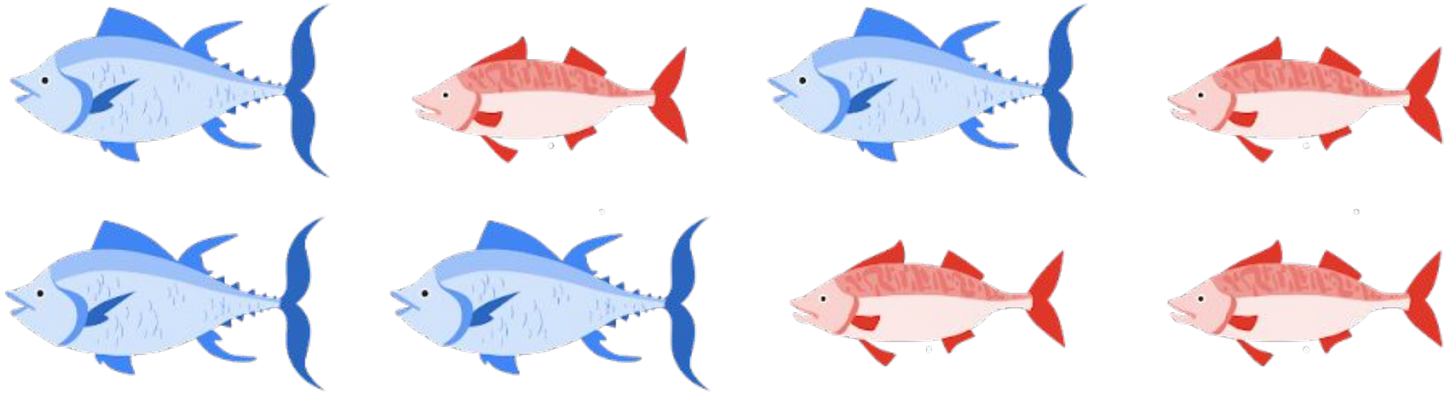
Total fish in the lake: 100
Caught: 80 fish + 80 rocks

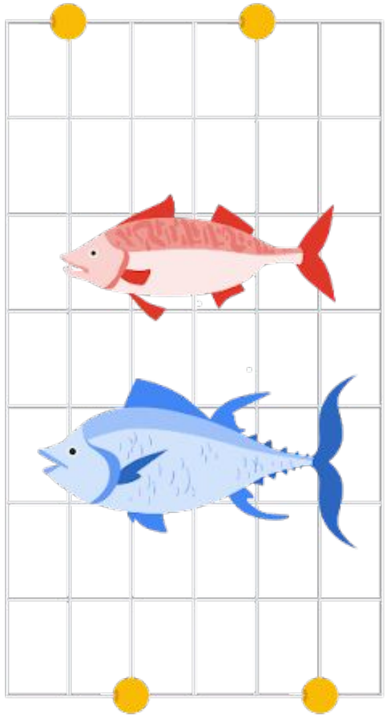
Recall: 80%

Precision: 50%

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Example: Fishing with a smaller net





Smaller net

Total fish in the lake: 100
Caught: 20 fish

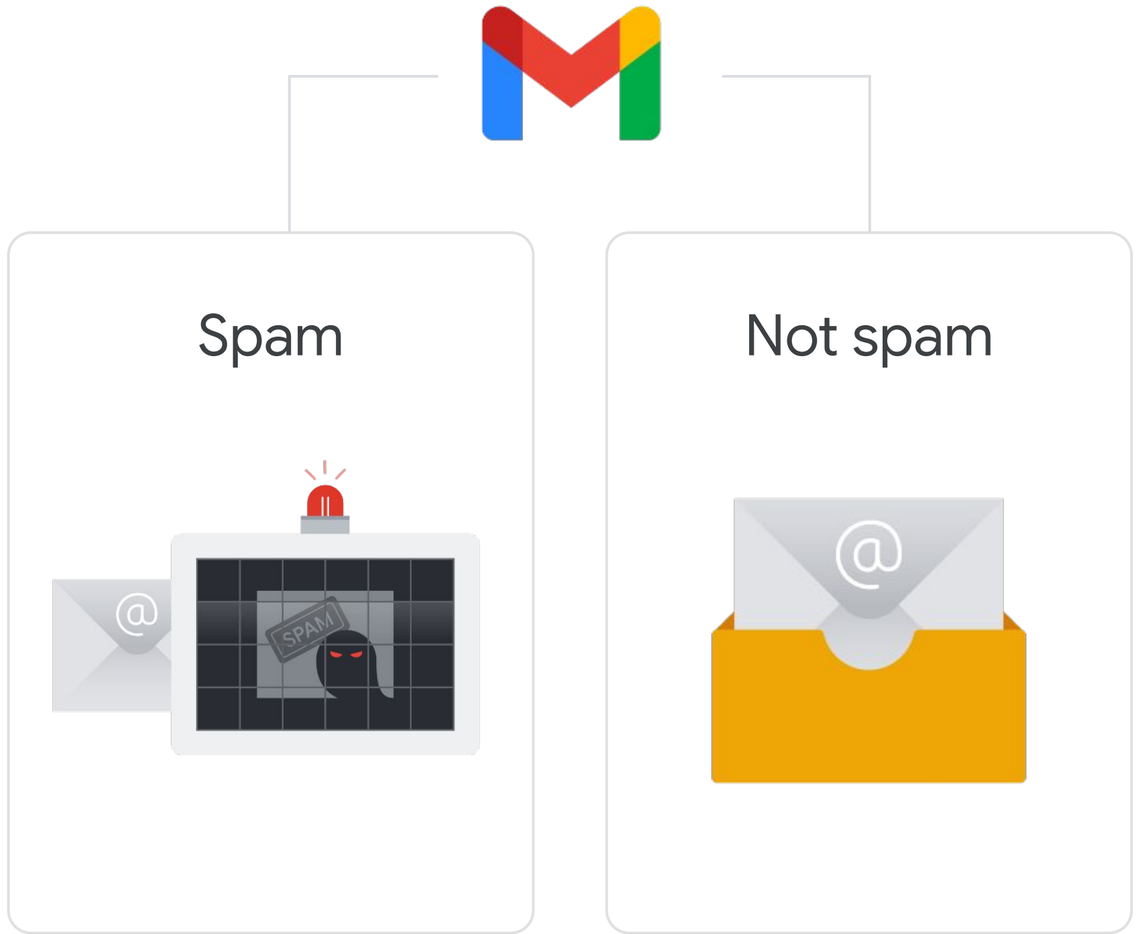
Recall: 20%

Precision: 100%

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Recall and precision: a trade-off

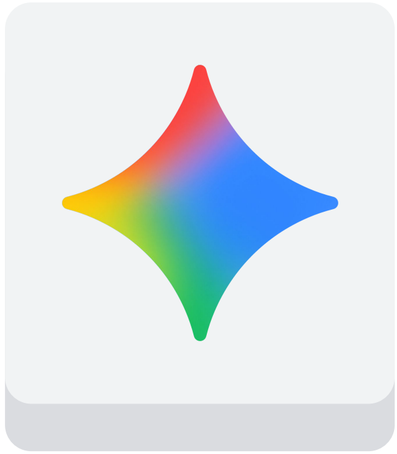
Recall



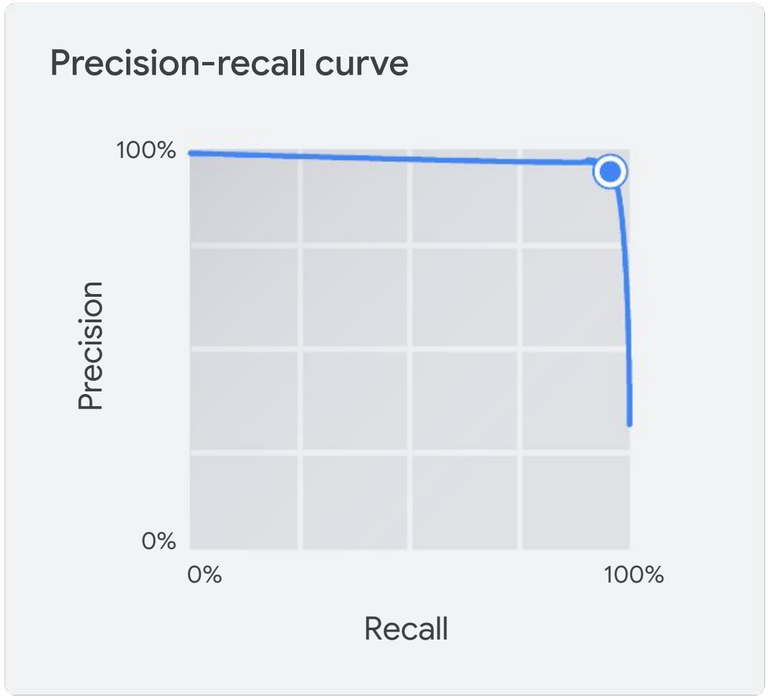
Precision

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The precision-recall curve



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Evaluation metrics: Feature importance

